

The Gantt Chart is

P0	P1	P2	P3	
0	5	8	16	22

Completion Time(CT): P0 =5, P1 =8, P2 = 16, P3 =22

Waiting Time and Turnaround Time of each process is as follows:

Process	Arrival Time (AT)	CPU burst Time (ms) (BT)	Turnaround Time (CT-AT)	Waiting Time (TAT-BT)
P0	0	5	5-0=5	5-5=0
P1	1	3	8-1=7	7-3=4
P2	2	8	16-2=14	14-8=6
P3	3	6	22-3=19	19-6=13

Average Turnaround time: $(5+7+14+19) / 4 = 11.25$ ms

Average Wait Time: $(0+4+6+13) / 4 = 5.75$ ms

(VII) Recourses Required

- Computer System with basic configuration. Linux/Ubuntu/ CentOS /any other open source Operating System.
- Software : Turbo C/ vi editor/ Any open source Python IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(VIII) Precautions to be followed

- (1) Handle computer system with care.
- (2) Strictly follow the instructions for writing, compiling, and executing the program.
- (3) Don't forget to save file before execution.
- (4) Follow safety Practices

(IX) Program Code

Consider the processes P1, P2, P3, P4 given in the below table, arrives for execution in the same order, with Arrival Time 0, and given Burst Time.

Implement First come first serve Scheduling algorithm to calculate find Average waiting time and Average turnaround time.

Process	Burst Time
P1	21
P2	6
P3	3
P4	2

Note : Attach the code at the end.

(This practical can be performed in any of the compiler like Turbo C/ vi editor etc or Python Interpreter / IDE like any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(X) Conclusion

Successfully executed python program to calculate Average Waiting Time and Turnaround Time of n processes with First Come First Serve (FCFS) CPU Scheduling Algorithm.

(X) Practical Related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. State the conditions for preemptive and non-preemptive scheduling algorithm.
2. State the disadvantages of FCFS.
3. Define Turnaround Time. Write a formula for average Turnaround Time.
4. Define Waiting Time. Write a formula for average Waiting Time.
5. The jobs are scheduled for execution as follows:

Solve the problem by using FCFS. Find Average waiting time and Average turnaround time using Gantt chart.

Process	Arrival Time	CPU burst Time (ms)
P0	0	10
P1	1	4
P2	2	14
P3	3	8

[Space to Write for Answers]

1) Preemptive : The CPU can be taken away from a running process before it completes.

Non-preemptive : Once a process gets the CPU, it continues until completion or it enters the waiting state.

2) 1. Poor average waiting time.

2. Convoy effect.

3. Short processes may wait behind long processes.

4. Not suitable for time-sharing systems.

3) Turnaround Time (TAT) : Time from process arrival to completion.

Formula : $TAT = \text{Completion Time} - \text{Arrival Time}$

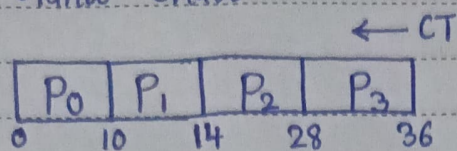
Average Turnaround Time : $\text{Avg TAT} = \frac{\text{Sum of all TAT}}{\text{Number of processes}}$

4) Waiting Time (WT) : Time a process spends waiting in the ready queue.

Formula : $WT = \text{Turnaround Time} - \text{Burst Time}$

Average Waiting Time : $\text{Avg WT} = \frac{\text{Sum of all WT}}{\text{Number of processes}}$

5) Gantt Chart



Process	AT	BT	CT	TAT	WT
P ₀	0	10	10	10	0
P ₁	1	4	14	13	9
P ₂	2	14	28	26	12
P ₃	3	8	36	33	25

Turnaround Time
(TAT = CT - AT)

$$P_0 = 10 - 0 = 10$$

$$P_1 = 14 - 1 = 13$$

$$P_2 = 28 - 2 = 26$$

$$P_3 = 36 - 3 = 33$$

Waiting Time
(WT = TAT - BT)

$$P_0 = 10 - 10 = 0$$

$$P_1 = 13 - 4 = 9$$

$$P_2 = 26 - 14 = 12$$

$$P_3 = 33 - 8 = 25$$

$$\text{Average Turnaround Time} = \frac{10 + 13 + 26 + 33}{4} = \frac{82}{4} = 20.5 \text{ units}$$

$$\text{Average Waiting Time} = \frac{0 + 9 + 12 + 25}{4} = \frac{46}{4} = 11.5 \text{ units}$$